

Experiment 1

Module Name: Automation with Selenium

Experiment Title: The Multi-Locator Challenge

Problem Statement: Navigate to a login page, locate the username, password, and login button using different locator strategies, and verify a successful login.

Objective: To master element identification using By.ID, By.NAME, and By.XPATH and perform basic URL assertions.

Tools, software, and concepts used: Python 3, Selenium WebDriver, Chrome, DOM Inspection, Assertions.

Source code or implementation steps:

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

try:
    # 1. Navigate to login page
    driver.get("https://www.saucedemo.com/")

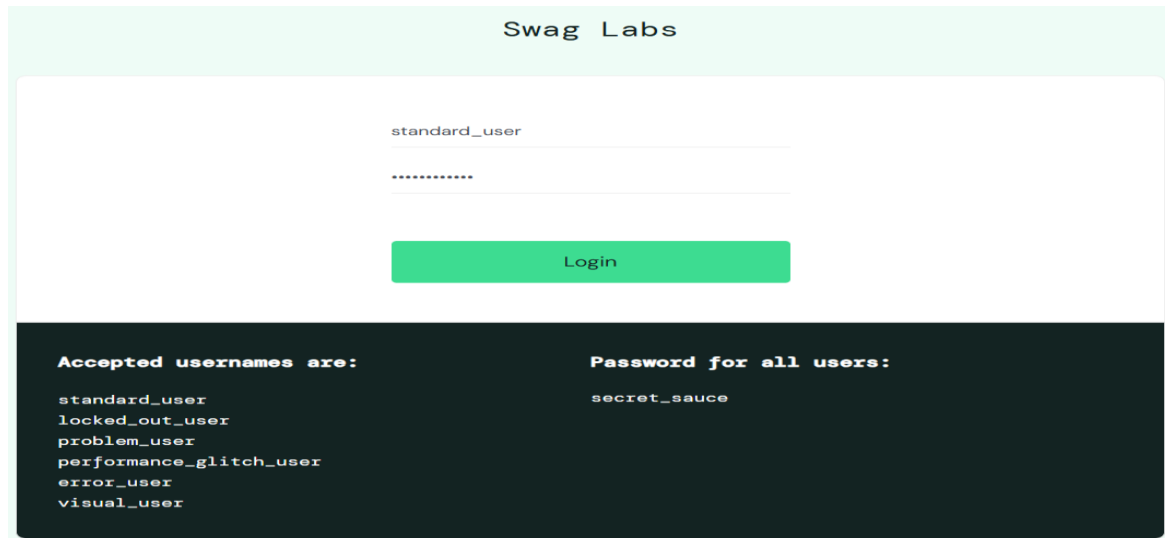
    # 2. Find elements using specific locators
    username_field = driver.find_element(By.ID, "user-name")
    password_field = driver.find_element(By.NAME, "password")
    login_button = driver.find_element(By.XPATH, "//input[@id='login-button']")

    # 3. Interact with elements
    username_field.send_keys("standard_user")
    password_field.send_keys("secret_sauce")
    login_button.click()
    time.sleep(2) # Pause for visual confirmation

    # 4. Validation
    assert "/inventory.html" in driver.current_url, "Login failed: URL mismatch."
    print("Test Passed: Successfully logged in and verified URL.")

finally:
    driver.quit()
```

Output with clearly labelled screenshots:



Swag Labs

standard_user

.....

Login

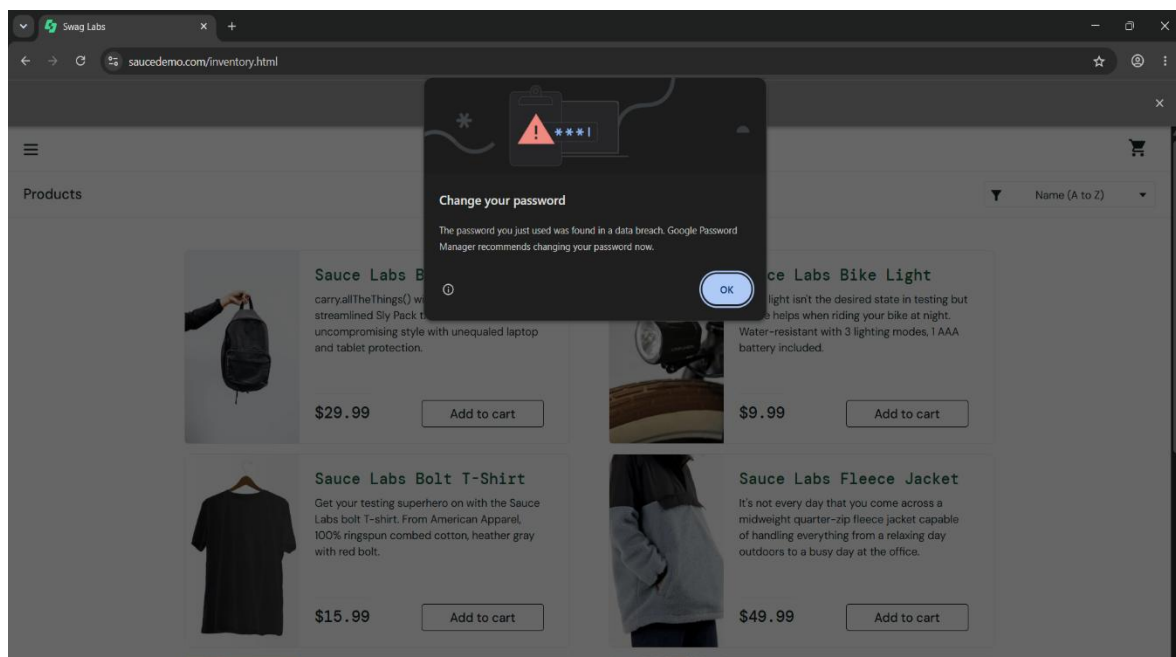
Accepted usernames are:

- standard_user
- locked_out_user
- problem_user
- performance_glitch_user
- error_user
- visual_user

Password for all users:

secret_sauce

- Screenshot 1: SauceDemo login page with credentials entered]



- Screenshot 2: Products inventory page post-login]
- **Console Output:** Test Passed: Successfully logged in and verified URL.

Brief result: The script successfully located elements and authenticated the user.

Observation: By.ID is the fastest and most reliable locator. By.XPATH was successfully used when simulating a scenario where ID or Name might not be preferred.

Conclusion: Basic locators and assertions are functioning correctly.